

Understanding BARCS

A self-study guide to the methodology, with worked examples,
a multiple-choice quiz, and an answer key

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1 The problem BARCS solves

A pooled CRISPR screen produces, for every guide g and every sequenced library i , a read count K_{gi} . Each library also has a total number of mapped guide reads, N_i . The question of interest is how guide abundance depends on the experimental design: time, an ordered FACS bin, a dose, a donor, a treatment, or an interaction between them.

CB² answered this for two groups. BARCS keeps the same sampling model but replaces the two group means with a regression on an arbitrary design matrix. Three objects travel together:

Object	Shape	Rule
counts	guides $\times$ libraries	raw integer read counts
totals	one per library	unfiltered mapped-guide totals N_i
data	one row per library	same order as the columns of counts

Key idea

BARCS treats a guide's count as a proportion of its library. The denominator N_i is the total recorded *before* any guide is filtered out, because removing guides and recomputing totals silently changes what every remaining coefficient means.

2 The sampling model

2.1 From binomial to beta-binomial

If every library of a given condition contained exactly the same guide proportion μ_i , sequencing would be pure binomial sampling: $K_{gi} \sim \text{Binomial}(N_i, \mu_i)$. Real libraries are biological replicates, so the true proportion itself varies from library to library. BARCS lets the per-library proportion P_i follow a beta distribution:

$$K_i \mid P_i \sim \text{Binomial}(N_i, P_i), \quad P_i \sim \text{Beta}(\mu_i \kappa, (1 - \mu_i) \kappa), \quad \rho = \frac{1}{\kappa + 1}. \quad (1)$$

Marginally, K_i is beta-binomial with mean $N_i \mu_i$ and variance

$$\text{Var}(K_i) = N_i \mu_i (1 - \mu_i) \{1 + (N_i - 1) \rho\}. \quad (2)$$

The intraclass correlation $\rho \geq 0$ is the overdispersion: $\rho = 0$ recovers the binomial.

2.2 Why this separates depth from replication

Divide by N_i to get the variance of the observed proportion $Y_i = K_i/N_i$:

$$\text{Var}(Y_i) = \mu_i (1 - \mu_i) \left\{ \frac{1 - \rho}{N_i} + \rho \right\}. \quad (3)$$

The first term shrinks as sequencing depth N_i grows; the second does not. Once a library is sequenced deeply, the remaining uncertainty is biological.

Worked example: deeper sequencing barely helps

Take $\mu = 10^{-4}$ and $\rho = 10^{-5}$. The bracket in Equation (3) is

$$N = 10^6 : 10^{-6} + 10^{-5} = 1.1 \times 10^{-5}, \quad N = 10^7 : 10^{-7} + 10^{-5} = 1.01 \times 10^{-5}.$$

Ten times more reads reduce the variance by only about 8%. Adding a second *library*, by contrast, halves the variance of a mean. This is the whole reason BARCS takes its degrees of freedom from libraries, not reads.

3 The regression model

For guide g in library i ,

$$K_{gi} \mid N_i \sim \text{BetaBinomial}(N_i, \mu_{gi}, \rho_g), \quad \text{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g. \quad (4)$$

Every guide has its own coefficients $\boldsymbol{\beta}_g$ and its own dispersion ρ_g . The design vector $\mathbf{x}_i$ is an ordinary row of an R model matrix, so `~ dose + batch`, `~ replicate + time`, or `~ knockout * treatment` all work.

Interpreting a coefficient. β_{gd} is the change in log-odds of the guide’s library share per unit of predictor d , holding the other columns fixed. Because guide proportions are tiny, $\text{logit}(\mu) \approx \log \mu$, so β_{gd} is approximately a log fold change in abundance per unit of x_d .

Contrasts. Any linear combination $\mathbf{c}^\top \boldsymbol{\beta}$ can be tested, for example the difference between two treatment arms (`bb_contrast()`).

4 Estimation: feasible IRLS

BARCS fits each guide separately by feasible iteratively reweighted least squares. Suppressing the guide index, with current $\boldsymbol{\beta}$ and ρ :

$$\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad \mu_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}}, \quad (5)$$

$$z_i = \eta_i + \frac{Y_i - \mu_i}{\mu_i(1 - \mu_i)}, \quad w_i = \frac{N_i \mu_i(1 - \mu_i)}{1 + (N_i - 1)\rho}. \quad (6)$$

The update is weighted least squares on the working response: $\hat{\boldsymbol{\beta}}_{\text{new}} = (X^\top W X)^{-1} X^\top W \mathbf{z}$.

Estimating ρ . At each fitted mean, ρ solves the Pearson estimating equation

$$\sum_{i=1}^m \frac{(K_i - N_i \hat{\mu}_i)^2}{N_i \hat{\mu}_i (1 - \hat{\mu}_i) \{1 + (N_i - 1) \hat{\rho}\}} = m - q, \quad (7)$$

where m is the number of libraries and q the number of coefficients. The left side decreases in ρ . Three cases arise:

1. If the binomial Pearson statistic ($\rho = 0$) is already $\leq m - q$, the counts are not overdispersed and $\hat{\rho}$ is truncated at 0.

2. Otherwise BARCS finds the root numerically.
3. If even ρ at its upper bound leaves the statistic above $m - q$, $\hat{\rho}$ is set to the bound and the covariance is multiplied by the remaining Pearson scale.

Coefficient and dispersion updates alternate until the coefficients change by less than the tolerance.

The weight saturates. For $\rho > 0$, $w_i \rightarrow \mu_i(1 - \mu_i)/\rho$ as $N_i \rightarrow \infty$. A library's weight cannot grow without bound with depth, which is Equation (3) seen from the fitting side.

In the package

`bbreg()` fits one guide; `bb_screen()` fits every row of the count matrix (optionally in parallel). The weighted cross-products are computed in C++ with ReppArmadillo.

5 Inference for a single guide

At convergence the covariance is $\widehat{\text{Cov}}(\hat{\beta}) = (X^\top \widehat{W} X)^{-1}$ and, for coefficient j ,

$$t_j = \frac{\hat{\beta}_j}{\sqrt{\widehat{\text{Cov}}_{jj}}} \sim t_{m-q}. \quad (8)$$

The reference degrees of freedom are $m - q$: independent libraries minus coefficients. Total read count never enters.

Worked example: degrees of freedom in the Cas13 screens

Each cell line has 6 libraries (days 0, 7, 14 $\times$ 2 replicates) and the model `~ replicate + time` has 3 coefficients (intercept, replicate, time). Each guide therefore has $6 - 3 = 3$ residual degrees of freedom. A variance estimated from 3 degrees of freedom is very noisy, which motivates the next section.

The t reference is heavier-tailed than a normal, but it treats $\hat{\rho}$ as known. It is not an exact correction for dispersion uncertainty; moderation and negative controls are the practical safeguards.

6 Dispersion moderation (on by default)

6.1 The idea

With 3 residual degrees of freedom, some guides look quiet by chance and get tiny standard errors, others look noisy and get huge ones. Moderation shrinks each guide's variance toward a library-wide trend, as `limma` and `edgeR` do.

6.2 The procedure

1. **Raw inflation.** Let $\nu = m - q$ and $s_g^2 = Q_g^{(0)}/\nu$, where $Q_g^{(0)}$ is the guide's Pearson statistic under the *binomial* model. Unlike $\hat{\rho}$, s_g^2 is not truncated, so it behaves like a scaled χ_ν^2/ν .
2. **Trend.** Fit a lowess curve of $\log s_g^2$ against $\log$ mean CPM (span 0.5). Call the fitted value $\log s_{\text{trend},g}^2$.

3. **Prior degrees of freedom (Smyth 2004).** Let $e_g = \log s_g^2 - \log s_{\text{trend},g}^2 - \psi(\nu/2) + \log(\nu/2)$, with ψ the digamma function. Compute

$$\text{excess} = \widehat{\text{Var}}(e_g) - \psi'(\nu/2).$$

If $\text{excess} > 0$, solve $\psi'(d_0/2) = \text{excess}$ for d_0 ; otherwise $d_0 = \infty$ (complete shrinkage).

4. **Prior scale.** $s_{0g}^2 = \exp\{\log s_{\text{trend},g}^2 + \bar{e} + \psi(d_0/2) - \log(d_0/2)\}$.

5. **Posterior.**

$$\tilde{s}_g^2 = \frac{\nu s_g^2 + d_0 s_{0g}^2}{\nu + d_0}. \quad (9)$$

6. **Rescale.** The fitted standard error already reflects $\max(1, s_g^2)$, so

$$\text{SE}_{g,\text{mod}} = \text{SE}_g \sqrt{\tilde{s}_g^2 / \max(1, s_g^2)}, \quad t_{g,\text{mod}} = \hat{\beta}_g / \text{SE}_{g,\text{mod}}, \quad \text{df} = \nu + d_0.$$

The coefficient $\hat{\beta}_g$ never changes; only its standard error and reference distribution do.

Worked example: one guide

$\nu = 3$, $s_g^2 = 4$, prior $s_{0g}^2 = 1.5$, $d_0 = 12$.

$$\tilde{s}_g^2 = \frac{3(4) + 12(1.5)}{15} = 2.0, \quad \text{rescale} = \sqrt{2/4} = 0.71, \quad \text{df} = 15.$$

The noisy guide's standard error shrinks by 29% and its test now uses a t_{15} reference instead of t_3 . A guide with $s_g^2 = 0.5$ would instead be pulled *up* toward the prior, protecting against false confidence.

In the package

`bb_screen(moderate = NULL)` moderates when at least 50 guides are usable; `moderate = FALSE` turns it off. Options: `trend = FALSE` shrinks to one constant; `one_way = TRUE` only ever raises a variance; `borrow_df = FALSE` keeps ν degrees of freedom. A result cannot be moderated twice.

Why it matters. In the Cas13 screens, moderation raised essential-gene average precision from 0.838 to 0.876, level with `limma-voom` and `edgeR-QL`, and improved every cell line.

7 The denominator: what a coefficient is relative to

7.1 Full-library totals and composition shift

With N_i as the denominator, β measures change in library share. If many guides deplete (a dropout screen), every surviving guide's share rises even when its absolute abundance is unchanged. Null guides then have positive coefficients, and their t statistics can be large.

7.2 Control-based totals

`barcs_control_totals()` holds a chosen control class C at a constant share of every library:

$$N_i^{\text{ctrl}} = \text{round}\left(\frac{\sum_{g \in C} K_{gi}}{\sum_{g \in C} K_g} \bar{N}\right), \quad \text{raised if needed so that } N_i^{\text{ctrl}} \geq \max_g K_{gi}. \quad (10)$$

Coefficients then describe abundance relative to the controls.

Worked example: control totals

Two libraries; two control guides with 500 reads each in both; one hit with 1,000 reads in library 1 and 100 in library 2. Full totals are 2,000 and 1,100 (mean 1,550). Control totals are $1000 \times 1550 / 1000 = 1550$ in both libraries, so the controls keep the same share and the hit's depletion is measured against them.

Choosing the control class. Non-targeting guides do not cut; safe-harbor guides cut but hit no gene. Normalizing on safe-harbor guides absorbs the shared cutting response. Guides used to build the totals should not also be used to judge the result.

8 Negative-control calibration

When the model's assumptions are violated (for example, FACS bins from one donor partition the same cells), null t statistics are too wide. `bb_calibrate_controls()` rescales all t statistics by one factor estimated from negative controls:

- `tail_quantile`: scale = $Q_{0.95}(|t_{\text{ctrl}}|) / t_{\nu, 0.975}$;
- `qq_slope`: the slope, through the origin, of control quantiles against t quantiles over the 0.50–0.95 band (less variable);

and then scale $\leftarrow \max(1, \text{scale})$, so calibration can only make tests more conservative. Standard errors are multiplied and t statistics divided by the scale; estimates are unchanged.

Worked example: tail scale

Controls' 95th percentile of $|t|$ is 3.5 with $\nu = 4$. Since $t_{4, 0.975} = 2.776$, the scale is $3.5 / 2.776 = 1.26$: every t is divided by 1.26.

Honest evaluation. If the same controls choose the scale and are then used to report the null rate, the 5% rate is guaranteed by construction. The IL2RA analysis therefore uses five-fold cross-fitting: each control is evaluated with a scale estimated from the other four folds.

9 From guides to genes

9.1 Directional Stouffer

For the j th valid guide of gene g , convert the two-sided p -value back to a signed normal score, then sum:

$$z_{gj} = \text{sign}(\hat{\beta}_{gj}) \Phi^{-1}\left(1 - \frac{p_{gj}}{2}\right), \quad Z_g = \frac{\sum_{j=1}^{m_g} z_{gj}}{\sqrt{m_g}}, \quad p_g = 2\Phi(-|Z_g|). \quad (11)$$

Concordant guides reinforce each other; discordant ones cancel. The reported gene effect is the *median* guide coefficient, which resists one extreme guide. Gene p -values are then BH-adjusted.

9.2 The independence problem

$\sqrt{m_g}$ is the standard deviation of a sum of m_g *independent* standard normals. But guides of one gene share every library; if they share noise, their null scores have positive correlation r , and

$$\text{Var}\left(\sum_j z_{gj}\right) = m_g + m_g(m_g - 1)r. \quad (12)$$

BARCS divides by the square root of this instead:

$$Z_g^c = \frac{\sum_j z_{gj}}{\sqrt{m_g + m_g(m_g - 1)\hat{r}}}. \quad (13)$$

Worked example: what correlation does

Five guides, each with $z = 2.5$, so $\sum z = 12.5$. Uncorrected, $Z = 12.5/\sqrt{5} = 5.59$. Suppose instead the five guides have sum $\sum z = 5.59$ (each $z \approx 1.12$). Uncorrected $Z = 2.5$, $p = 0.012$. With $r = 0.4$, the variance is $5 + 5 \cdot 4 \cdot 0.4 = 13$ and $Z^c = 5.59/\sqrt{13} = 1.55$, $p = 0.12$: no longer significant.

9.3 Estimating r without being fooled by real signal

1. Take each guide's Pearson residual vector across libraries and scale it to unit length.
2. For each gene with at least two guides, average the pairwise inner products (correlations); then average over genes, giving each gene equal weight (so a large control "gene" cannot dominate).
3. Do the same for deterministic pairs of guides from *different* genes; this baseline removes correlation shared by the whole library.
4. $\hat{r} = \min\{1, \max(0, \text{within} - \text{between})\}$.

Key idea

Residuals are what is left after the fitted mean has absorbed every design effect, including a true gene effect. A gene whose guides all deplete strongly therefore does *not* have correlated residuals; only noise shared by its guides does. That is why the correction does not penalize real hits.

In simulations with $r = 0.4$, the corrected gene-level type I error was 0.018–0.052 versus up to 0.262 uncorrected. In the real screens analyzed, $\hat{r}$ was 0.008–0.026, so the correction changed little there.

9.4 Experimental alternatives

Four other summaries pool effect sizes rather than tests: `bb_gene_normal()`, `bb_gene_consistency()`, `bb_gene_partial_pool()`, and `bb_gene_eb_moderate()`. They ranked genes worse than Stouffer in the simulated FACS benchmarks and are kept as sensitivity analyses.

10 How the benchmarks measure success

Metric	Meaning
AUROC	Probability a random positive outranks a random negative.
Average precision (AP)	Mean precision at each true positive's rank; sensitive to the top of the list.
Recall at 5% null FPR	Fraction of positives above the score that only 5% of null genes exceed; threshold-free with respect to calibration.
Recall at FDR 0.10	Fraction of positives called at the method's own nominal FDR; depends on calibration.
Realized FDP	Fraction of calls that are false, when truth is known.
Proxy null	Genes assumed null (e.g. unexpressed lncRNAs); imperfect, so null rates are diagnostics.

Headline results (BARCS 0.2.0).

Benchmark	BARCS	Comparison
Cas13 AP	0.876	limma 0.876, edgeR 0.874, MAGeCK 0.877
Cas13 recall @ FDR 0.10	0.767	0.692–0.767
IL2RA validated regulators	23/26, 61 calls	MAGeCK 17/26, 72 calls
CRISPulator F1 (moderated)	0.847	unmoderated 0.811
Null grid, correlated guides	0.018–0.052	uncorrected up to 0.262
A375 recall @ FDR 0.05	0.883	MAGeCK 0.710

11 Limitations to keep in mind

- **Independent libraries.** FACS bins from one donor and repeated harvests from one culture are dependent. Donor indicators adjust means, and controls fix the miscalibration, but the dependence itself is not modeled.
- **Plug-in dispersion.** The t reference treats $\hat{\rho}$ as known; moderation reduces, but does not remove, this approximation.
- **Likelihood-compatible counts.** The Cas13 benchmark used normalized, batch-corrected values rounded to integers; a raw-count rerun would be the stronger test.
- **Denominator choice.** Full-library and control-based coefficients answer different questions; choose before looking at results.

12 Pipeline at a glance

1. `barcs_quant()`: FASTQ $\rightarrow$ counts, totals.
2. Optional `barcs_control_totals()`: choose the reference.
3. `bb_screen()`: per-guide beta-binomial regression, t tests, dispersion moderation, residual correlation $\hat{r}$.
4. Optional `bb_calibrate_controls()`: negative-control scale.

5. `bb_gene_stouffer()`: correlation-aware gene statistic, median effect, BH FDR.

13 Quiz

Choose the single best answer. The answer key with explanations follows on a separate page.

- Q1.** In BARCS, the total N_i for library i should be
- (A) the median count across guides
 - (B) the sum of counts after removing low-count guides
 - (C) the full mapped-guide total recorded before any guide filtering
 - (D) the number of FASTQ reads in the file
- Q2.** The beta-binomial variance of a guide count is $N\mu(1-\mu)\{1+(N-1)\rho\}$. Setting $\rho = 0$ gives
- (A) zero variance
 - (B) a binomial variance
 - (C) a Poisson variance
 - (D) a negative-binomial variance
- Q3.** With $\mu = 10^{-4}$ and $\rho = 10^{-5}$, increasing depth from 10^6 to 10^7 reads reduces the variance of the observed proportion by roughly
- (A) 8%
 - (B) 50%
 - (C) 90%
 - (D) 0%
- Q4.** The residual degrees of freedom of a BARCS guide test are
- (A) number of independent libraries minus number of coefficients
 - (B) number of guides per gene minus one
 - (C) fixed at 30
 - (D) total reads minus number of coefficients
- Q5.** In the model $\text{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g$, a coefficient β_{gd} is best interpreted as
- (A) the overdispersion of the guide
 - (B) the probability that the guide is essential
 - (C) the change in raw read count per unit of x_d
 - (D) approximately the log fold change in guide abundance per unit of x_d , holding other columns fixed
- Q6.** As $N_i \rightarrow \infty$ with $\rho > 0$, the IRLS weight $w_i = N_i\mu_i(1-\mu_i)/\{1+(N_i-1)\rho\}$
- (A) grows without bound
 - (B) approaches zero
 - (C) approaches $\mu_i(1-\mu_i)/\rho$
 - (D) equals N_i

- Q7.** If a guide's binomial Pearson statistic is already below $m - q$, BARCS sets $\hat{\rho}$ to
- (A) 0
 - (B) the library-wide median
 - (C) $1/(m - q)$
 - (D) its upper bound
- Q8.** A Cas13 cell line has 6 libraries and the model $\sim \text{replicate} + \text{time}$. How many residual degrees of freedom does each guide have?
- (A) 5
 - (B) 3
 - (C) 4
 - (D) 2
- Q9.** Dispersion moderation changes
- (A) the library totals
 - (B) the standard error and the reference degrees of freedom, but not $\hat{\beta}_g$
 - (C) the guide-to-gene assignment
 - (D) the coefficient estimate $\hat{\beta}_g$
- Q10.** Moderation shrinks each guide's log variance inflation toward
- (A) a lowess trend in log mean CPM across all guides
 - (B) zero
 - (C) the inflation of the negative controls only
 - (D) the largest inflation in the library
- Q11.** The prior degrees of freedom d_0 are estimated by
- (A) a fixed constant of 4
 - (B) maximum likelihood on the counts
 - (C) cross-validation on essential genes
 - (D) the scaled- F moment method of Smyth (2004), using digamma and trigamma functions
- Q12.** With $\nu = 3$, $s_g^2 = 4$, $s_{0g}^2 = 1.5$, $d_0 = 12$, the moderated inflation $\tilde{s}_g^2$ is
- (A) 2.75
 - (B) 4.0
 - (C) 2.0
 - (D) 1.5
- Q13.** Why is moderation based on $s_g^2 = Q_g^{(0)}/\nu$ rather than on $\hat{\rho}_g$?
- (A) $\hat{\rho}_g$ depends on read depth only
 - (B) s_g^2 ignores the design
 - (C) $\hat{\rho}_g$ is not stored by `bb.screen()`

(D) $\hat{\rho}_g$ is truncated at zero, so it does not behave like a scaled χ^2 quantity

Q14. By default, `bb_screen()` moderates dispersions when

- (A) the user supplies negative controls
- (B) the design has more than two groups
- (C) at least 50 guides are usable
- (D) always, even with 5 guides

Q15. In a dropout screen with full-library totals, null guides tend to show

- (A) negative coefficients, because all guides deplete
- (B) undefined coefficients
- (C) coefficients centered at zero
- (D) positive coefficients, because surviving guides gain library share

Q16. `barcs_control_totals()` produces totals that

- (A) are normalized to counts per million
- (B) hold a chosen control class at a constant share of every library
- (C) exclude the control guides
- (D) equal the number of FASTQ reads

Q17. Normalizing on safe-harbor rather than non-targeting guides

- (A) absorbs the shared cutting response into the reference
- (B) removes all composition shift and all cutting effects
- (C) makes the totals non-integer
- (D) has no effect

Q18. `bb_calibrate_controls()` with `min_scale = 1` can

- (A) only make tests more liberal
- (B) only make tests more conservative (or leave them unchanged)
- (C) change the gene assignment
- (D) change coefficient estimates

Q19. Control 95th percentile $|t| = 3.5$ with $\nu = 4$ ($t_{4,0.975} = 2.776$). The tail scale is about

- (A) 3.50
- (B) 1.00
- (C) 1.26
- (D) 0.79

Q20. Why did the IL2RA analysis use five-fold cross-fitting of control calibration?

- (A) Because the screen has five donors
- (B) To speed up computation

- (C) Because estimating the scale and evaluating the null rate on the same controls makes a 5% rate true by construction
- (D) To estimate guide correlation

Q21. In directional Stouffer, a guide's signed score is

- (A) $-\log_{10} p_{gj}$
- (B) $\hat{\beta}_{gj}/\text{SE}_{gj}$ always
- (C) $\text{sign}(\hat{\beta}_{gj}) \Phi^{-1}(1 - p_{gj}/2)$
- (D) $\Phi^{-1}(p_{gj})$

Q22. The reported BARCS gene effect is

- (A) the mean guide coefficient
- (B) the median guide coefficient
- (C) the Stouffer Z
- (D) the largest guide coefficient

Q23. If m null guide scores each have variance 1 and pairwise correlation r , the variance of their sum is

- (A) m
- (B) $m + m(m - 1)r$
- (C) $m^2 r$
- (D) $m(1 - r)$

Q24. With 5 guides and $r = 0.4$, the corrected Stouffer denominator is $\sqrt{13}$ instead of $\sqrt{5}$. A raw $Z = 2.5$ ($p = 0.012$) becomes about

- (A) $Z = 4.0, p < 0.001$
- (B) $Z = 1.55, p = 0.12$
- (C) $Z = 2.5, p = 0.012$
- (D) $Z = 0, p = 1$

Q25. BARCS estimates within-gene correlation from residuals because

- (A) a true gene effect is absorbed by the fitted mean, so it does not inflate residual correlation, whereas shared noise does
- (B) residuals do not depend on the design
- (C) residuals are always independent
- (D) residuals are smaller than counts

Q26. Why does BARCS subtract the average correlation of guides from *different* genes?

- (A) To account for guide length
- (B) To make $\hat{r}$ negative
- (C) Because different genes are perfectly correlated

- (D) To remove correlation shared by the whole library, such as normalization artifacts
- Q27.** In the within-gene average, each gene receives equal weight so that
- (A) the estimate is always zero
 - (B) genes with many guides dominate
 - (C) essential genes are ignored
 - (D) a large group of control guides labelled as one “gene” cannot dominate the estimate
- Q28.** In the real screens analyzed, the estimated within-gene correlation was
- (A) 0.008–0.026, so the correction changed little
 - (B) larger than 0.5 in every screen
 - (C) about 0.4
 - (D) exactly zero by definition
- Q29.** Recall at a 5% empirical null false-positive rate differs from recall at FDR 0.10 because it
- (A) ignores ranking
 - (B) requires negative controls
 - (C) uses only essential genes
 - (D) does not depend on how well the method’s p -values are calibrated
- Q30.** In the Cas13 benchmark, the main reason BARCS caught up with limma and edgeR in version 0.2.0 was
- (A) removing the day-7 libraries
 - (B) switching to a negative-binomial likelihood
 - (C) using control-based totals
 - (D) dispersion moderation borrowing strength across guides
- Q31.** Which statement about FACS bins from one donor is correct?
- (A) They should be summed before analysis
 - (B) They require a binomial model
 - (C) They partition the same cells and are dependent; donor indicators adjust means but do not model that dependence
 - (D) They are independent libraries, so BARCS models them exactly
- Q32.** The experimental gene summaries (`bb_gene_partial_pool()`, etc.) are kept because
- (A) they are required by `bb_screen()`
 - (B) they replace calibration
 - (C) they ranked genes best in the benchmarks
 - (D) they serve as sensitivity analyses; Stouffer ranked genes better in the simulated FACS benchmarks

14 Answer key with explanations

- Q1. C.** The denominator must describe the library that was sequenced. Recomputing it after filtering silently changes every remaining guide's share (Section 1).
- Q2. B.** With $\rho = 0$ the bracket is 1, leaving $N\mu(1 - \mu)$, the binomial variance.
- Q3. A.** The bracket $(1 - \rho)/N + \rho$ goes from 1.1×10^{-5} to 1.01×10^{-5} , about an 8% reduction: biological variation dominates.
- Q4. A.** $m - q$. Depth affects precision within a library but adds no degrees of freedom.
- Q5. D.** Guide proportions are tiny, so log-odds $\approx$ log proportion; the coefficient is a conditional log fold change.
- Q6. C.** The denominator grows like $N\rho$, so the weight saturates at $\mu(1 - \mu)/\rho$.
- Q7. A.** No overdispersion beyond binomial is detectable, so $\hat{\rho}$ is truncated at 0.
- Q8. B.** $6 - 3 = 3$.
- Q9. B.** Moderation rescales the standard error and adds prior degrees of freedom; $\hat{\beta}_g$ is unchanged.
- Q10. A.** The prior follows a lowess trend of log inflation against log mean CPM (or a constant with `trend = FALSE`).
- Q11. D.** The excess variance of the centered log inflations over $\psi'(\nu/2)$ is matched to $\psi'(d_0/2)$.
- Q12. C.** $(3 \cdot 4 + 12 \cdot 1.5)/15 = 30/15 = 2.0$.
- Q13. D.** Truncation at zero distorts the distribution; the untruncated Pearson inflation behaves like a scaled χ^2_ν/ν , which the scaled- F model assumes.
- Q14. C.** With fewer than 50 usable guides the prior cannot be estimated reliably, so moderation is skipped unless forced.
- Q15. D.** Composition shift: as essential guides drop out, survivors gain share, so null coefficients shift positive.
- Q16. B.** Control totals keep the control class at a fixed share, rescaled to the mean library size and rounded.
- Q17. A.** Safe-harbor guides cut DNA, so normalizing on them removes the shared cutting response from targeted effects.
- Q18. B.** The scale is floored at 1, so test statistics can only be divided by a number ≥ 1 .
- Q19. C.** $3.5/2.776 = 1.26$.
- Q20. C.** A held-out evaluation avoids the circularity of judging calibration on the controls that set it.
- Q21. C.** The two-sided p is mapped back to a normal deviate and given the coefficient's sign.
- Q22. B.** The median resists a single extreme guide.

- Q23. B.** m variance terms plus $m(m - 1)$ covariance terms of size r .
- Q24. B.** $Z^c = 2.5\sqrt{5}/\sqrt{13} = 1.55$, $p = 0.12$.
- Q25. A.** Residuals exclude fitted design effects, so only shared noise produces residual correlation.
- Q26. D.** The between-gene baseline removes library-wide correlation that is not specific to a gene.
- Q27. D.** Equal gene weighting prevents a large pseudo-gene from dominating the average.
- Q28. A.** Estimated $\hat{r}$ was 0.008–0.026 (Cas13), 0.011 (IL2RA), and 0.011 (Evers).
- Q29. D.** It fixes the false-positive rate empirically using proxy nulls, so it compares ranking independent of p -value calibration.
- Q30. D.** Moderation raised AP from 0.838 to 0.876; without it each guide’s variance rests on only 3 degrees of freedom.
- Q31. C.** Bins partition one cell pool and are negatively dependent; negative-control calibration corrects the resulting miscalibration, and Waterbear models the dependence directly.
- Q32. D.** They are retained for sensitivity analysis; Stouffer gave higher average precision and F1 in most simulated scenarios.